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Power grid vulnerability analysis using complex network theory: A topological study of the Nordic transmission grid

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ABSTRACT

To reduce the vulnerability of power grids to high impact low probability (HILP) events, analysis methods can be applied to quantify the criticality of the nodes in the grid. The method implemented in this article is one originating from complex network theory. It is used to quantify the structural vulnerability of an open-source transmission grid model representing the Nordic transmission grid. The analytical measures used are clustering coefficient and betweenness, closeness, degree, and combined centrality, which are weighted with respect to the estimated values of the transmission lines' series reactance. The results, which are presented in the form of geographic and network representations, show substantial differences in terms of criticality between the nodes. The most critical ones are highlighted in geographic representations and are further compared with an open-source system analysis performed by the Swedish transmission system operator (TSO). The outcome from this study is that the weighted and combined centrality measure performed the best in terms of identifying critical nodes in the Nordic transmission grid. Thus, the method can be used as a tool for assessing the structural vulnerability of a real transmission grid, even with limited access to electrical grid data. However, the results from this method should not be considered conclusive.

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1. Introduction

The survival of modern societies is dependent on critical infrastructures [1–3]. One of these critical infrastructures is the electrical power system, having a determinant role in the economy, security, safety, health, and well-being of communities [3]. This makes the electrical power grid an infrastructure where high resilience and low vulnerability are desirable.

However, during recent years, the number of interruptions in the power system caused by extreme weather events and targeted attacks have increased rapidly. The effects of large but rare events, so called high impact low probability (HILP) events, have led to major blackouts and substantial damage to the power grid worldwide [4]. In 2008, an ice storm hit China causing blackout for 200 million people with a direct cost of 2.2 billion USD [5]. In August 2013, the superstorm Sandy caused extensive power outages in the US, leaving 8.5 million customers without electricity [6]. The storm Gudrun hit the Scandinavian countries in January 2005, causing power outage for more than 660 thousand customers and damaged over 30 000 kilometres overhead lines. Some of these customers were out of electricity for as long as 45 days [7]. Hence, to reduce the consequences for the power system caused by HILP events, knowledge about the vulnerable parts of the

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system is desired. Thereby, finding tools for analysing the vulnerability of power grids is a research field of high interest for engineers in both industry as well as academia.

Consequently, between 2016 and 2018, the Swedish transmission system operator (TSO) Svenska kraftnät performed a major system analysis of the Nordic transmission grid, focusing more specifically on the Swedish part of the grid [8]. The aim of the analysis was to identify how to expand the transmission grid to increase transmission capacity, decrease losses, facilitate for transmission line outages, and to make the grid topology more robust. Among other, the analysis resulted in the biggest investment project in the history of Svenska kraftnät, called NordSyd, in which major investments in the transmission grid are planned. The locations of the new transmission line terminals constitute of already existing substations, six northwards and six southwards, respectively [8].

1.1. Related work

In previous research, vulnerability has been defined in multiple ways, focusing on different perspectives of the concept [3]. The vulnerability of the Nordic transmission grid has been investigated in [9], aiming at identifying the scenarios, situations and incidents leading to serious consequences for the power system. In [10], vulnerability is defined as "an expression for the problems a system faces to maintain its function if a threat leads to an unwanted event and the problems the system faces to resume its activities after the event occurred". Another definition from the literature is "the weakness level of a system to failures, disasters or attacks" [11]. A fourth definition, specified on power grids is given by [12], stating that "robustness or vulnerability (its opposite concept) are often used to measure to what extent a power grid has high or low reliability, respectively". Based on these definitions, vulnerability is defined in this study as the measure to which extent the nodes in a network are critical to the network's structure and functionality.

Even though the practical applicability of complex network methods in the context of power grid analysis is open to debate [1,3,13,14], these methods have reached a significant scientific impact, used extensively in power grid research [2,15–37]. Within this area of research, two major approaches have been formed: a purely topological and a hybrid approach [12].

In [22], the Nordic transmission grid as well as a synthetic representation of the western USA transmission grid were studied with respect to the structural vulnerability of the systems, using a complex network theory approach. The grid consisted of undirected and unweighted edges which represented the physical structure of the power grid. Thus, the study is purely topological without any electrical dimensions taken into consideration. The paper presents results in terms of degree distributions, average clustering coefficient and average path length of the systems. It also examines the error and attack tolerance of the different grid models. Further, [26] examined the robustness of two power grid models, IEEE300 and a randomly generated 1000-node model. The study applied a combination of analytics, ranging from a purely topological perspective to an artificial flow model as well as a DC power flow model. The study showed that a combination of different approaches seems promising when evaluating the structural robustness of a power grid model.

Furthermore, in [12] and [38], the pure topological measures have been adjusted to fit the examination of electrical power systems. In these studies, the resistance and impedance of the cables and overhead lines have been used as the weighting factors, respectively. The vulnerability of the IEEE-30 and IEEE-118 models were assessed in [39] by applying complex network concepts where the edges were weighted with respect to the power lines' reactance values. In [40], the structural vulnerability of the Indian transmission grid together with three generic IEEE models were investigated by applying both a weighted and an unweighted approach aiming to identify critical buses and transmission lines. For these grid models, a weighted approach gave more realistic results than the unweighted approach. The betweenness and netability were investigated for the IEEE-14 model in [33] in order to exhibit the critical lines in the model. To weight the edges with parameters capturing the physical properties of a power grid have been used extensively in the field of research.

1.2. Motivation, aim and structure of paper

Even though many studies have used complex network theory to perform power grid analyses, most studies have focused on either test grid models or grid models of real systems which data is protected by secrecy. In this study, a power grid model, fully based on open-source data, representing a simplification of the Nordic synchronous transmission grid is analysed with respect to its structural vulnerability. Thus, geographical presentations of the nodes' criticality are made possible. This way of presenting results as well as applying complex network concepts has, to our knowledge, never been done before. Also, a new method for handling transformers when converting the model's buses into network nodes is applied. Furthermore, this study proposes a new measure based on a heuristic approach, combining three well-established centrality measures.

The study aims to examine to what degree the Nordic transmission grid is tightly interconnected, and which nodes in the simplified grid model that are indicated to be the most critical from a complex network theory perspective. Further, a comparison between a purely topological, unweighted approach and a hybrid, weighted approach is done to determine whether the selected weighting factor has a substantial impact on the results or not and to investigate if a weighted approach performs better than an unweighted approach in terms of finding critical nodes in a power grid.

Comparisons with state-of-the-art methods, such as sophisticated simulation software, is possible when having a complete grid model containing all data necessary. However, to get valuable results from such simulations, access to

reliable data is necessary. Such data could be the generators' and the loads' characteristics together with high resolution time series data, detailed information about the transmission lines properties, and protection relay configurations. Since most data concerning the Nordic transmission grid is protected by secrecy, reliable results from such sophisticated simulations are not possible to achieve outside the TSOs. Thus, the evaluation of the proposed method must be done differently from conventional quantitative methods. The results are therefore compared qualitatively with the open-access system analysis presented by Svenska kraftnät in [8]. This system analysis is considered as the best source of public information with respect to the vulnerability of the Nordic transmission grid.

The main contributions of the paper can be summarized as follows:

- Assessment of the structural vulnerability of a simplified grid model representing a real transmission grid by applying complex network concepts and using open-source data.
- Qualitative comparison between the study's results and a power system analysis published by one of the responsible TSOs.
- A new approach on how to handle transformers in a grid model when converting the model's buses into network nodes.
- A new centrality measure based on a heuristic approach, combining three established centrality measures.

This paper is structured as follows: Section 2 presents the methodology and data of the study including network theory fundamentals, analytical measures, review of the case object as well as an illustration of the methodological workflow. Section 3 covers the results of the study, including clustering coefficient and multiple centrality scores. In Section 4, the results are discussed and in Section 5 the study is concluded.

2. Methodology and data

2.1. Complex network theory fundamentals

Complex network theory has been used extensively in research to study different types of real world networks such as the Internet, road infrastructure as well as power systems [2,3,5,12,16–27,41–44]. The fundamental components within complex network theory are a graph G containing a set of vertices V, a set of edges E with a corresponding set of weights W, G = {V, E, W} [27]. In the context of this paper, the graph is a mathematical representation of the physical power grid where the vertices represent the substations, and the edges represent the transmission lines. The vertex v_i can be identified by its index i. Similarly, the edge connecting vertex i to vertex j is identified by the notation e_{ij} with corresponding weight w_{ij}.

Furthermore, graphs are divided into two subgroups: directed and undirected graphs [12]. A directed graph has edges associated with a direction, that is, a flow can only pass through the edge in one predefined direction. However, for undirected graphs, the flow can go in any direction. Since direction is an aspect that has no relevance in the context of this study, all graphs observed have been considered as undirected graphs. Since this study has examined a power grid, the terminology 'network', 'node' and 'link' are frequently used when referring to the mathematical terminology 'graph', 'vertex' and 'edge'.

2.2. Analytical measures

To quantify the structural vulnerability of the Nordic transmission grid, clustering coefficient and four centrality measures have been used. In contrast with [16] that uses five centrality measures, three centrality measures have been adopted in this study together with a fourth measure that is a combination of the three first-mentioned centrality measures.

The centrality measures used in this study derivate from a purely topological approach. However, previous research has indicated that a purely topological approach can give misleading results since it does not take the fundamental electrical properties in consideration [15,27,43]. Thus, both a purely topological approach as well as a hybrid approach have been adopted in this study.

2.2.1. Path and step length

The average path length l is a measure used to calculate the average distance between any pair of nodes in the network [12,22]. It is defined as

$$l = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij} \tag{1}$$

where N is the number of nodes and d_{ij} is the shortest distance between node i and j. The higher the value of the average path length is, the more scattered the network is relatively to the number on nodes [22]. Average path length is a common measure when comparing the degree of scattering in different kinds of networks [12,22]. However, since this study focuses

on one single network, the average path length has not been used. Yet, because of its relationship with the path length measure, it should be mentioned for the sake of completeness.

In its simplest mathematical form, the distance d between two neighbouring nodes equals to 1. However, in this study the distance is measured in two ways. The first way, which is used in the purely topological approach is the simple mathematical form, with d_{ij} equals to 1 for any neighbouring pair of nodes i and j. The second way, which is used in the hybrid approach, defines the distance d_{ij} between node i and j as a variable dependent on the line series reactance X_{ij}. The distances according to the two different approaches can be formulated as

$$d_{ij} = \begin{cases} n_{ij}, & \textit{purely topological approach} \\ \left|X_{ij}\right|, & \textit{hybrid approach} \end{cases} \tag{2}$$

where n_{ij} is the number of steps along the shortest path between node i and j, and $|X_{ij}|$ is the absolute value of the transmission line series reactance along the shortest path between node i and j, measured in ohm. The reactance values are calculated as

$$X_{ij} = x_{ij}\left(U_{ij}\right) \cdot s \tag{3}$$

where x_{ij} is the line series reactance per length unit, defined as a function of the voltage level of the transmission line going from node i to node j, U_{ij}, and s is the physical length of the transmission line. The physical lengths of the transmission lines have been calculated using the nodes' longitude and latitude coordinates given by [45]. The series reactance values are standard values for overhead lines and are obtained from [45].

The absolute value of the series reactance has been chosen as the length parameter since the series reactance is multiple times larger than the resistance, and thus, it is the dominating part of the line impedance. This is done in contrast with other studies, for example [12,37,38,40], where the line resistance or impedance is used as length parameter. Furthermore, the absolute value of the transmission line series reactance is affected by the length as well as by the physical construction of the transmission line. Thereby, transmission line voltage level has indirectly also been considered.

In an ideal case, more transmission line data such as current magnitudes and power flows could be used as weighting factors. Also, weighting of the nodes' importance could be of interest. However, none of these are open-source data. Further, one aim of this study is to investigate whether it is possible to assess the structural vulnerability of a real power grid based on a limited amount of open-source data. Therefore, a straightforward weighting factor based on line reactance is chosen for the purpose of this study.

2.2.2. Clustering coefficient

Clustering coefficient is a measure capturing the density of two nodes connected to each other and at the same time directly connected to a common third node. This connectivity structure can be compared to triangles in a network, making the network robust and tightly interconnected [12]. Thus, the clustering coefficient can be interpreted as a measure that describes the degree of meshing in a grid. The local clustering coefficient C_i for node i in an undirected graph is defined as

$$C_i = \frac{2\left|\left\{e_{jk} : v_j, v_k \in N_i, e_{jk} \in E\right\}\right|}{k_i(k_i - 1)} = \frac{2M_i}{k_i\,(k_i - 1)} \tag{4}$$

where e_{jk} is an edge in the set of edges E, v_i and v_k are nodes, N_i is the neighbourhood for node i and k_i is the number of neighbours connected to node i. M_i is thereby representing the number of links that connect these k_i nodes to each other. Fig. 1 shows examples of node interconnections and what clustering coefficients these connections correspond to.

The average clustering coefficient C is based on the local property clustering coefficient and can be calculated as

$$C = \frac{1}{N}\sum_i C_i \tag{5}$$

where C is the mean value of each individual clustering coefficient [18]. The higher the density of triangles in the network structure is, the higher the average clustering coefficient will be. Thus, the closer C gets to 1, the more tightly interconnected and robust the network structure is. Consequently, clustering coefficient does not consider the distance between nodes but is solely dependent on the structure of interconnections between them. Clustering coefficient can be used to compare the degree of clustering between multiple networks. However, it can also be used when analysing a single network. In particular, the distribution of the clustering coefficient values is valuable to get a concise overview of to what extent the network is clustered. Thus, the clustering coefficient complements the perspectives captured by the centrality measures defined in Section 2.2.3 by giving an overarching description of the meshing of the network. Therefore, it is of relevance for this study.

2.2.3. Centrality measures

Centrality can be obtained as a function $C(v_i)$ where $C(v_i) > C(v_j)$ implicates that node v_i is more critical than node v_j if and only if the condition is fulfilled [44]. In this study, four centrality measures have been adopted whereof three are well-established in the existing literature and one is new. The three well-established centrality measures are

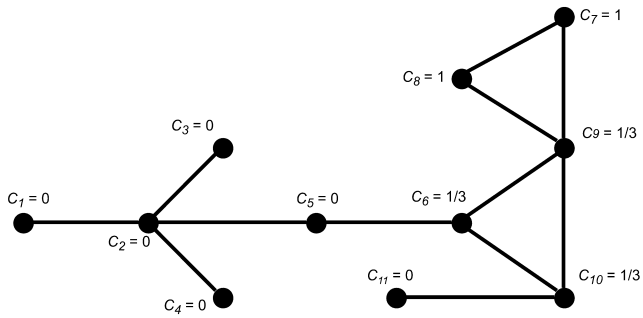

Fig. 1. Examples of node interconnections with corresponding clustering coefficient values.

betweenness, degree, and closeness centrality [32]. These measures are proposed since they capture three important, yet completely different dimensions that are of significance in a power system network [46]. Furthermore, these three measures are intuitive and used extensively in complex network research [32]. Therefore, it is of interest to investigate if a combination of these three centrality measures can be more powerful in terms of finding possibly critical nodes than the three individual measures. Thus, the fourth centrality measure proposed in this study is a combination of the three aforementioned measures, called combined centrality.

Betweenness centrality $C_B(v)$ quantifies how often a node v is found in the shortest path between two nodes s and t in a network [12,16]. Since the graphs in this study are undirected, the path from s to t and from t to s is counted only once. However, since there can be more than one shortest path between s and t, the betweenness centrality is calculated as

$$C_B(v) = \sum_{s \neq t \neq v \in V} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{6}$$

where σ_{st} is the total numbers of shortest paths between s and t, of which $\sigma_{st}(v)$ number of shortest paths pass through node v. The length of the path is determined using Eq. (2). Betweenness centrality is a measure specific for each node, and it indicates how critical node v is in terms of node interconnection. If node v has a high C_B score, a failure of node v will lead to multiple node pairs to either be connected via longer paths or disconnected from each other.

The degree centrality $C_D(v)$ of a node v is a measure that can be interpreted in terms of how much a node v influences its neighbouring nodes and links if the node gets deactivated or removed [12,16]. It is a measure of how many links that are connected to node v and is defined as

$$C_D(v) = \sum_{i=1}^{N} A_{vi} \tag{7}$$

where $\{v, i, N\} \in V$ and A_{vi} is the number of direct interconnections between node v and node i. For a simple graph, A is the same as the adjacency matrix.

The closeness centrality $C_C(v)$ for node v can be used to find out how close a node is to all other nodes [46]. The higher score of $C_C(v)$, the closer node v is to the other nodes in the network and the more critical the node is for the network to be closely interconnected. Closeness centrality is defined as

$$C_C(v) = \frac{N-1}{\sum_i d_{vi}} \tag{8}$$

where $\{v, i, N\} \in V$ and d_{vi} is the distance from node v to node i. The distances are calculated using Eq. (2).

However, even though centrality measures have been shown to be useful tools for identifying the critical elements in a network, previous research has indicated only a vague correlation between the different centrality measures themselves [16]. Therefore, in line with previous research [16], a heuristic strategy has been adopted where the different centrality measures have been combined. In this study, we propose a straightforward and equally weighted method for combining the centrality measures as

$$C_{comb}(v) = \frac{\frac{C_B(v)}{\sum_i C_B(i)} + \frac{C_C(v)}{\sum_i C_C(i)} + \frac{C_D(v)}{\sum_i C_D(i)}}{3} \tag{9}$$

where $C_{comb}(v)$ is the combined centrality for node v.

When all nodes have been assigned their respective centrality value, the nodes are grouped into 10 to 30 bins ranging from the lowest to the highest centrality score. This procedure has been performed for all four centrality measures, and the number of bins has been adapted depending on the spread of the specific centrality measure. The nodes categorized in the highest bin for each centrality measure have been highlighted with red circles in the geographic plots as shown in Section 3.

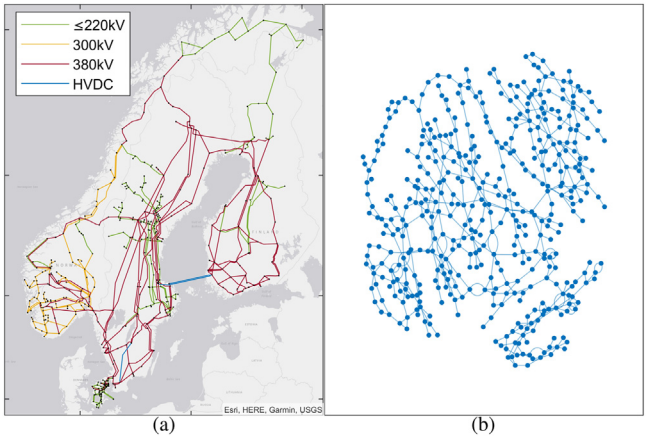

Fig. 2. (a) Geographic and (b) network representation of the adjusted Nordic490 model.

2.3. Power grid model

In this study, the Nordic490 transmission grid model has been used as test case. The Nordic490 model, first published in [45], is an open-source transmission grid model representing the Nordic transmission grid, including the transmission grids in Norway, Sweden, Finland, and eastern Denmark. The model is an extended and improved variant of the PyPSA-Eur open model, which is an open-source transmission grid model representing the European transmission grid [47]. The Nordic490 model consists of 494 busses, 600 AC transmission lines and 56 transformers distributed over voltage levels of 132 kV up to 380 kV. Originally, the model was created with the purpose of being an open-source power flow model for solving AC and DC power flow problems. However, its structure with buses and transmission lines can be converted into a network representation that can be analysed though complex network theory. Thus, the Nordic490 model is a useful tool for analysing the structural vulnerability of the Nordic transmission grid. The geographic and network representations of the Nordic490 model is shown in Fig. 2.

The Swedish transmission grid is characterised by an oblong grid structure, transferring the electricity over long distances from the hydropower plants in the north to the cities in the south. Also, the middle part of Sweden is characterised by a high density of wind farms, connected to the transmission grid. Norway has a high share of hydro power connected to the grid meanwhile Finland has several nuclear power plants serving the Finish transmission grid. Denmark is dominated by a high share of wind power production.

There are multiple HVDC links connecting the Nordic transmission grid to countries outside of the Nordic states. These connections join the Nordic transmission grid to the UK, the Netherlands, western Denmark, Germany, Poland, and the Baltic states. HVDC links within the Nordic transmission grid are Sydvästlänken which is located in the southern part of Sweden, Fenno-Skan which are two links between Sweden and Finland, and links connecting the islands Gotland and Åland to the main grid.

However, adjustments and assumptions have been made to the Nordic490 model to make it more well adapted to the purpose of this study. Firstly, HVDC links not used for interconnection of the Nordic synchronous AC grid have been removed. This include HVDC links to countries outside of the Nordic region as well as HVDC links to Åland, Gotland and Jylland. These HVDC links are removed since they are connected to buses geographically outside the Nordic synchronous transmission grid. HVDC links still in use in the revised model are Sydvästlänken and Fenno-Skan. Secondly, the HVDC links have been replaced by 380 kV AC lines. This is motivated by the fact that the HVDC links used in the revised model are connected between 380 kV AC lines and have similar properties in terms of length and transmission capacity as the 380 kV AC lines. Thirdly, all lines in the model representing transformers have been removed. Thus, each node pair i and j representing the primary and secondary side of a transformer have been replaced with a single node i. See Section 2.4 for details about this methodological strategy. Fourthly, the underground cables have been assumed to have the same properties as the overhead lines. This is motivated by the negligible number of underground cables in the Nordic transmission grid.

2.4. Transformer removal strategy

In the Nordic490 model, as well as in most grid models, transformers are mathematically modelled as transmission lines from node i to node j. Consequently, when edge e_{ij} is modelling a transformer, this implies that node i and j are situated at the same geographical position. However, in this study, we propose a new method for handling the transformers when applying complex network concepts. We argue that the results can be misleading when the

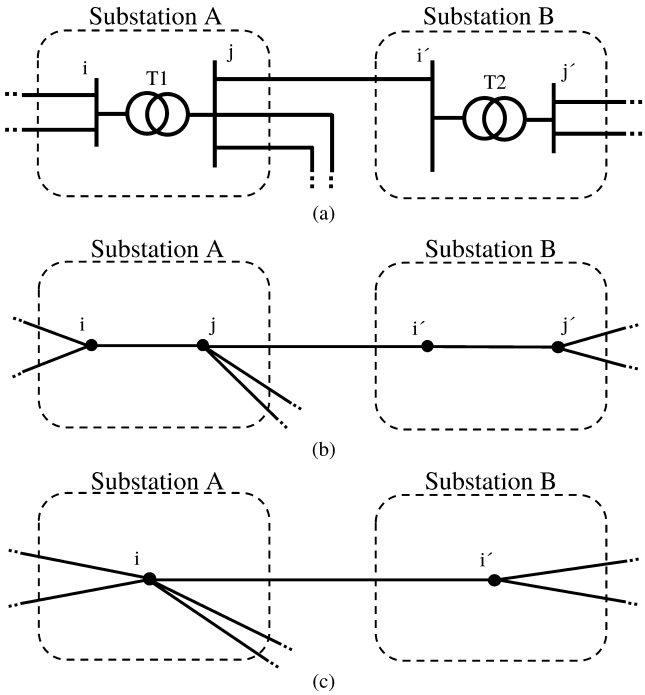

Fig. 3. Illustration of the transformer removal strategy applied on a subnet of two substations containing one transformer each. (a) One line diagram of the subnet. (b) and (c) show network representations without and with the transformer removal strategy applied, respectively.

Table 1
Network properties with and without the transformer removal strategy applied.

	Without transformer removal strategy	With transformer removal strategy	Intended physical interpretation
degree(i)	3	5	Number of transmission lines connected to Substation A.
degree(i')	2	3	Number of transmission lines connected to Substation B.
$d_{ii'}$	2	1	Unweighted distance between Substation A and B.

transformers are kept in the model, if the aim of the study is to quantify which substations pose the most critical for the system. Therefore, we propose a method where all the individual lines are checked whether they are transmission lines or transformers. For all lines identified as transformers, the secondary side's nodes, labelled *j* in Substation A in Fig. 3(a), are removed. Then, the lines themselves identified as transformers are removed. Finally, all lines originally connected to node *j* are reconnected to node *i*. This method is referred to as the transformer removal strategy. By applying the transformer removal strategy, every single substation is represented by a single node with transmission lines connecting the substation to other substations. To the authors knowledge, this approach has not been used in previous similar studies.

From Fig. 3(a), the single-line diagram of a subnet consisting of substation A and B is shown. When converting the single line diagram to a network representation, the transformers become edges as shown in Fig. 3(b). As can be obtained from the figure, problem emerges with multiple nodes representing one single substation, resulting in misleading results. Thus, the transformers, modelled as transmission lines between node *i* and *j* are removed and replaced by a single node *i*. Some of the differences between the network properties with and without the transformer removal strategy applied are shown in Table 1. Moreover, the betweenness and closeness centrality are also affected by the transformer removal strategy, even though they are not explicitly illustrated in Table 1.

2.5. Methodological workflow

From Fig. 4, a schematic flowchart representing the methodological workflow of the study is shown. Firstly, the data used in the study was extracted from the Nordic490 model. The data, containing information about the system's buses, transformers, transmission lines as well as geographical location of all parts of the model, was loaded into a MATLAB

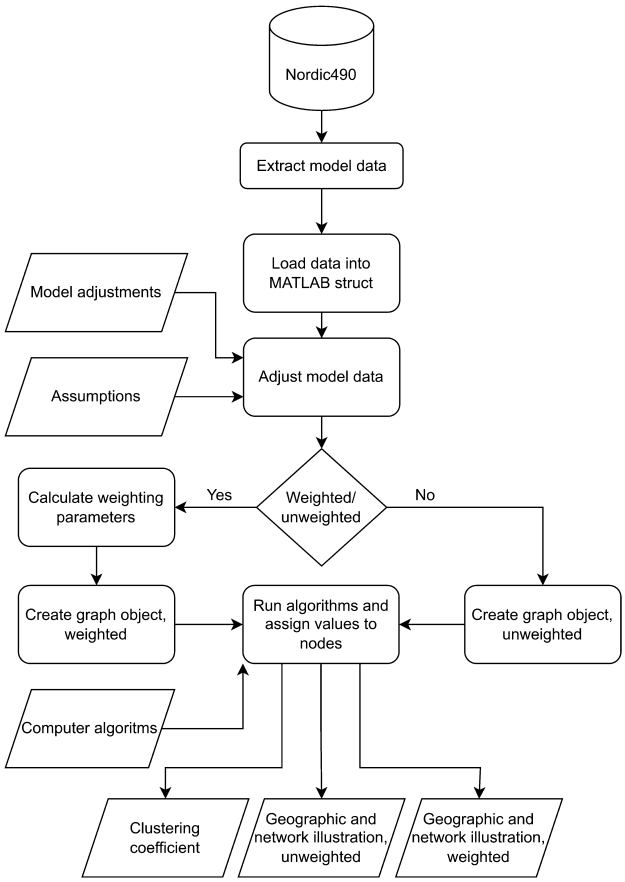

Fig. 4. Methodological workflow of the study.

struct. Then, the data was adjusted to fit the study and assumptions were made according to Section 2.3. A weighted as well as an unweighted graph object were created, and the calculation algorithms were applied to the graphs. This resulted in multiple graphs and measures, presented according to the outputs shown in Fig. 4.

3. Results

3.1. Clustering coefficient

In Fig. 5, the probability density function of the clustering coefficients for the Nordic490 model is shown. The average clustering coefficient $C = 0.0548$. From the figure, it can be seen that 84% of the nodes in the model have a clustering coefficient $C_i = 0$. These nodes are placed either at the end of an outer branch or between two or more nodes that are not directly interconnected to each other.

However, the dominating group of nodes with a clustering coefficient above zero is the group with $C_i = 1/3$, corresponding to 7% of all the nodes. These nodes are placed in triangular formations as illustrated in Fig. 1. Furthermore, the nodes with $C_i = 1$ are even more clustered relatively to their connection to other nodes, covering approximately 2% of the total number of nodes. About 0.5% of the nodes have a clustering coefficient score $1/3 < C_i < 1$. Compared with analyses of other European power grids, the Nordic power grid is loosely interconnected with respect to the clustering coefficient [48]. For example, the Spanish 400 kV network has an average clustering coefficient $C = 0.316$, the Italian 380 kV network $C = 0.156$, and the French 400 kV network $C = 0.279$ [48].

3.2. Betweenness centrality scores

Nodes with high betweenness centrality are often passed through when taking the shortest path from any node s to any node t. The critical nodes with respect to betweenness centrality are geographically located as shown in Fig. 6(a) and 6(c), highlighted with red circles. These nodes are placed in two areas: in the south-eastern part of Norway bordering Sweden and in the middle part of Sweden.

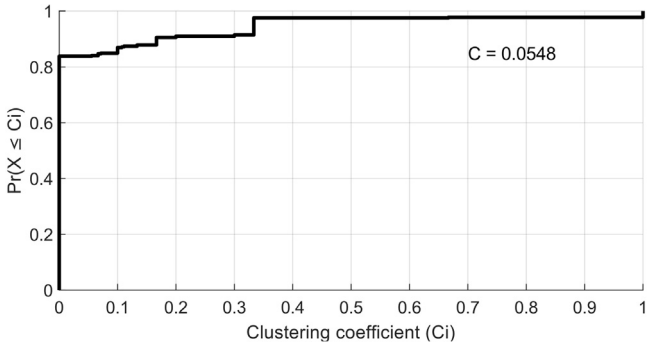

Fig. 5. Probability density function of clustering coefficients.

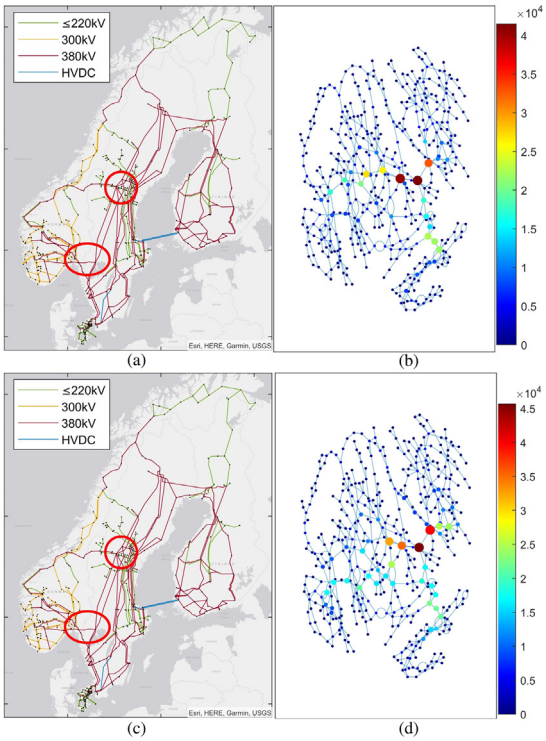

Fig. 6. (a, b) Unweighted and (c, d) weighted geographic and network representation of the betweenness centrality scores, Nordic490 model.

As shown in Fig. 6(b) and 6(d), the Nordic490 grid model is characterised by many low scored nodes in terms of betweenness centrality. Most of the nodes have a centrality score in the bottom of the score range. However, there are a handful number of nodes that have a significantly higher score, placed in the centre of the graph. The betweenness centrality measure shows a minor difference between the weighted and unweighted case.

3.3. Degree centrality scores

The geographic as well as the network representations of the Nordic490 model, with respect to weighted and unweighted degree centrality, are shown in Fig. 7. The geographic representations of the degree centrality scores, presented in Fig. 7(a) and 7(c), highlight ten critical nodes in the Nordic transmission grid. In Norway, the area around Oslo is marked as critical. Furthermore, two nodes nearby Copenhagen in Denmark and four nodes in Sweden have high degree centrality scores, indicating that these nodes are critical. Moreover, three nodes in Finland are to be considered as critical from a degree centrality point of view.

From the network representations shown in Fig. 7(b) and 7(d), the ten critical nodes can be observed. These nodes stand out significantly in terms of degree centrality, marked with nodes coloured in shades of red or yellow, respectively.

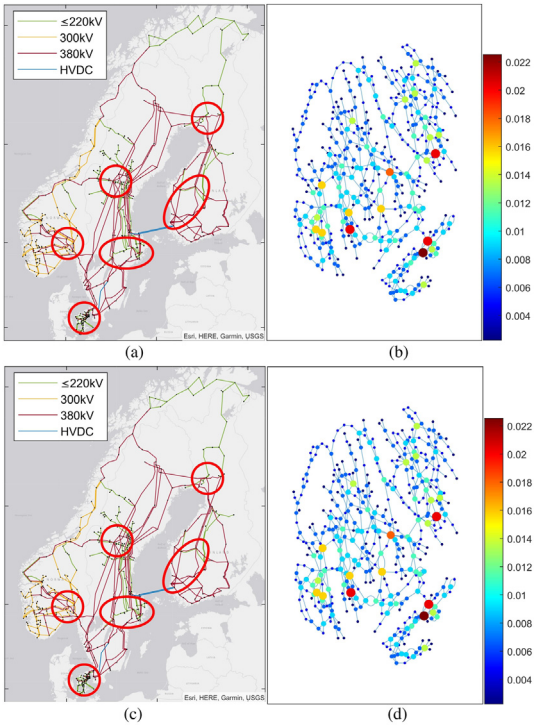

Fig. 7. (a, b) Unweighted and (c, d) weighted geographic and network representation of the degree centrality scores, Nordic490 model.

3.4. Closeness centrality scores

The closeness centrality scores of the Nordic490 grid model are shown in Fig. 8, presented as a geographic as well as a network representation. As can be seen from Fig. 8(a) and 8(c), there is a high concentration of critical nodes in the southern and middle part of Sweden. There are no highly scored nodes in Finland, Denmark, or Norway. From Fig. 8(b) and 8(d), it can be seen that the weighted case has a centrality score more evenly spread out in the network compared to the unweighted case.

3.5. Combined centrality scores

In Fig. 9, the results from the unweighted as well as the weighted combined centrality scores are shown. From the geographic representations shown in Fig. 9(a) and 9(c), the locations of the most critical nodes can be seen, highlighted with red circles. These nodes are in the middle part of Sweden as well as in the area between Oslo in Norway and Stockholm in Sweden. Moreover, one critical node is situated in Denmark. However, most of the nodes have a similar and relatively low combined centrality score. The weighted measure indicates more nodes as critical compared to the unweighted measure.

According to the network representation, a dozen nodes placed in the centre of the network stands out, marked with red, orange, and yellow colour, as shown in Fig. 9(b) and 9(d). Beside the high scored nodes, most of the nodes are low scored, marked with small, blue dots in the network representations.

3.6. Distributions of centrality scores

The distributions of the four different centrality measures are shown in Fig. 10. As indicated in the network and geographic representations shown in the previous sections, the distributions are similar or even identical between the weighted and the unweighted centrality scores. This accounts for all cases except from the closeness centrality score.

The betweenness centrality score follows an exponential curve with a high number of nodes reaching the scores at the bottom of the scores' range. However, there are still a few nodes reaching scores that are multiple times higher than the scores of most of the nodes. Furthermore, the degree centrality scores show a more spread-out distribution than the betweenness centrality distribution. Thus, the relative size of the criticality of the nodes is more evenly distributed with respect to the degree centrality than for the betweenness centrality. However, which is also shown in Fig. 7, there are still a few nodes with significantly higher centrality scores compared to the majority of the nodes' scores.

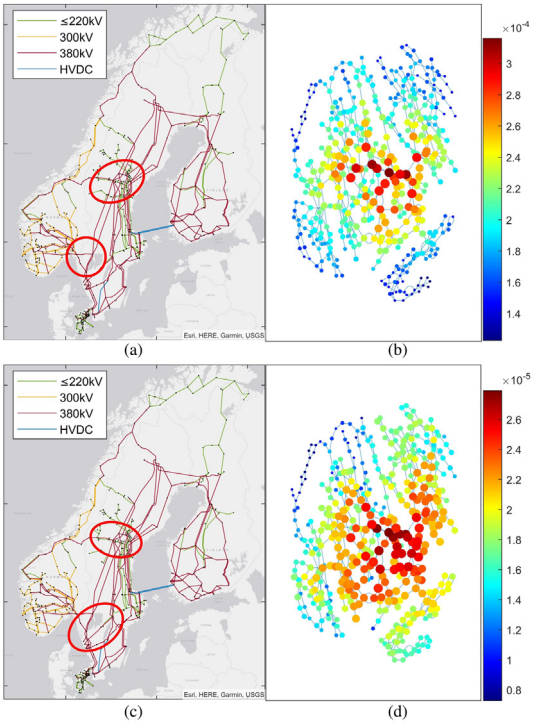

Fig. 8. (a, b) Unweighted and (c, d) weighted geographic and network representation of the closeness centrality scores, Nordic490 model.

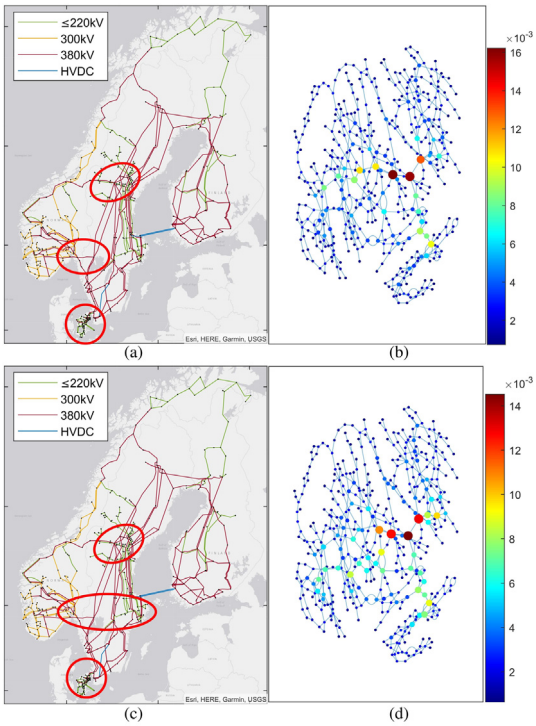

Fig. 9. (a, b) Unweighted and (c, d) weighted geographic and network representation of the combined centrality scores, Nordic 490 model.

From Fig. 10(c) it can be seen that the closeness centrality scores generate a narrow width with high amplitude for the weighted case and a wide width with a low amplitude for the unweighted case. The offset between the weighted and

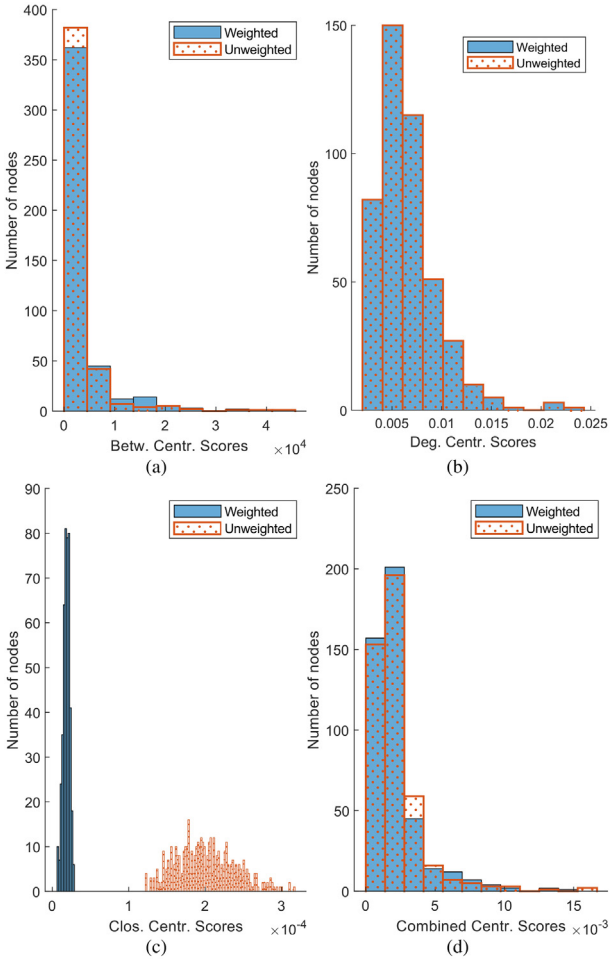

Fig. 10. Distributions with respect to: (a) betweenness, (b) degree, (c) closeness, and (d) combined centrality scores.

unweighted closeness centrality distributions is substantial. Finally, the distributions of the combined centrality scores are shown in Fig. 10(d). From the figure, it can be observed that most of the nodes have a low combined centrality score, which is in line with the results presented in Fig. 9. The relative difference between the weighted and unweighted case is small. However, a few nodes have a combined centrality score significantly larger than most of the nodes, which is slightly more common in the weighted case than in the unweighted case.

4. Discussion

4.1. Clustering coefficient and centrality measures

From this study, it is seen that the Nordic transmission grid is more loosely interconnected than other European transmission grids, which partly can be explained by the countries' geographical structures. Spain and France are more quadratic than Italy and the Nordic countries with respect to their structures, which is one reason explaining the high clustering coefficients of the Spanish and the French power grids. In contrast with the Italian power system, the power production facilities in Sweden and Norway are mostly located far from the consumers. This leads to an even more stretched out and loosely interconnected power grid, which is the reason why the Nordic power grid has a lower clustering coefficient than the Italian power grid. However, the clustering coefficient is a measure suitable for comparisons between networks rather than a standalone measure. Therefore, a comparison with a benchmark model would be suitable when evaluating the clustering coefficient distribution further.

When considering the geographic representations in Fig. 6(a) and 6(c), it can be observed that the critical nodes with respect to betweenness centrality are placed in areas where the Swedish and Norwegian transmission grids get interconnected. Consequently, the results are reasonable since the removal of any of the nodes important for the connection between Sweden and Norway will force the electrical power to flow significantly longer paths to get from

Sweden to Norway, and vice versa. With respect to betweenness centrality, the difference between the weighted and unweighted case was small. Thus, the betweenness centrality measure was not affected substantially by the weighting of the links in the network.

From Fig. 7, it can be concluded that the weighted and unweighted cases of the degree centrality are identical. This is, however, not surprising since the degree centrality measure only takes the number of nodes connected to node v as well as the total number of nodes in the network into account. In contrast with the betweenness centrality measure, there are relatively many highly scored nodes with respect to the degree centrality. Thus, the node criticality in terms of degree centrality is more spread out than in the case of betweenness centrality. The overlap between critical nodes with respect to betweenness and degree centrality is vague but still exists.

Closeness centrality was shown to be the centrality measure affected the most by the weighting factor of the links, see Fig. 8. Furthermore, with respect to the measure's relative size, the nodes' centrality measures are more evenly spread out in the case of weighted links compared to unweighted links. Thus, in the unweighted case, there are fewer nodes with high closeness centrality score than in the weighted case. Therefore, an unweighted case is more sensitive to node failures than a weighted case is, with respect to the closeness centrality score. However, the overall analysis of the closeness centrality measure, for both the weighted and unweighted case, is that the values span smaller ranges compared to the values of the betweenness and degree centrality measures. Thus, the network is to be considered as robust from a closeness centrality point of view.

From the three different centrality measures betweenness, degree and closeness centrality, information about the criticality of the nodes from three different perspectives is gained. By combining these measures into the combined centrality measure, an indication of which nodes are the most critical in the network from a centrality perspective is received. The results show a significant difference in terms of criticality for the nodes in the Nordic490 grid model, where most of the nodes in the grid have similar criticality. However, there are a dozen nodes in the model that pose significantly critical hubs in the network. It appears intuitive that most of the nodes found to be critical from a topological perspective are located in Sweden, since the Swedish transmission grid is the link between the other Nordic transmission grids.

With respect to the combined centrality, the weighted case identified more nodes than the unweighted case did in terms of criticality. Thus, the weighted approach seems to be more sensitive than the unweighted approach when identifying critical nodes.

From the centrality distributions shown in Fig. 10, it can be observed that betweenness, degree and combined centrality follows similar patterns. However, the distributions of the closeness centrality scores differ significantly between the weighted and the unweighted case as expected. Considering Eq. (8), the determining difference between the weighted and unweighted closeness centrality case is the way the distances between the nodes are calculated. Since $d_{ij,weighted} >> d_{ij,unweighted}$ for 99.7% of all links between neighbouring nodes in the Nordic490 model, and since $C_C(v)$ decreases when the distance d_{ij} increases, the weighted case has a more narrow distribution of closeness centrality scores than the unweighted case has.

4.2. Comparison with Svenska kraftnät's system analysis

The purposes of the NordSyd project, as presented by Svenska kraftnät, are to increase transmission capacity, decrease losses, facilitate for transmission line outages, and to make the grid topology more robust [8]. One of the driving forces is the transmission capacity limitations which have a negative impact on the reliability of supply for the southern part of Sweden [8]. Thus, one of the outcomes from the NordSyd project is a more robust transmission grid and thereby a reduced vulnerability. From the system analysis performed by Svenska kraftnät, six substations in the middle of Sweden as well as six substations in the southern of Sweden have been identified as objects in the transmission grid where major investments will be done, hereafter called NordSyd substations. Thus, it is reasonable to assume that these substations pose critical nodes in the transmission grid where the transmission capacity must be increased. Even though the criticality of specific substations in a grid are not openly available, it is argued that the criticality of the substations can be conjectured based on the abovementioned reasoning.

All four centrality measures were able to identify some of the NordSyd substations, but the weighted and combined centrality measure was the best measure in finding these substations. A total of five out of twelve NordSyd substations were identified with the proposed model. The model successfully identified nodes geographically located in the regions where all the NordSyd substations are placed. However, it should be emphasised that even though the model is able to find some nodes indicating to be critical, it does not imply that all critical nodes have been found. The results from this method should therefore not be considered conclusive.

By comparing the results from the present study and the system analysis performed by Svenska kraftnät, it can be concluded that the correlation between the results is substantial. However, the results might not be enough to be considered as a proof since they are more indirectly related to what is presented by Svenska kraftnät. Yet, the results strengthen the hypothesis that a complex network approach is relevant when identifying the critical nodes in a real transmission grid. This applies even though simplifications have been made and that the model is based solely on open-source data. This implies that a complex network approach based on open-source data can generate relevant and grounded results when searching for critical nodes in real transmission grids. However, most probably TSOs have their own methods to identify critical substations in a grid. These methods are based on data which are not openly available. Thereby, the

applications of complex network concepts in power grid analyses are more of interest for academic researchers not having access to all data needed to perform conventional analyses. Furthermore, this study does not investigate whether reactance is the best weighting factor or not. Thus, comparing different weighting factors is left for future research.

It should be emphasised that the Nordic490 model is a simplification of the real Nordic transmission grid, and that the topology is based on open-source data without any sensitive information. For example, the transmission line parameters are based on standard overhead line parameters for different voltage levels, the number of nodes is reduced in the model compared with the real grid, and the electrical characteristics of the loads are unknown. Thus, the specific results in relation to the actual Nordic transmission grid should be taken with caution. Furthermore, it can be assumed that the model does not give a complete picture of the vulnerability of the power grid since the input data is limited by secrecy. Moreover, the models can be improved by taking even more indicators such as operational constraints, frequency regulation, or the random and dynamic characteristics of the power grid into consideration. However, this is beyond the scope of the present study and is left for future studies. Still, the study indicates that complex network concepts can be used as tools when quantifying the structural vulnerability of a model representing a real transmission grid. Furthermore, based solely on open-source data, the specific results should be considered as a reasonable assessment of the structural vulnerability of the Nordic transmission grid.

Proposals for future research would be to quantify the vulnerability in terms of load flow analyses with production data, load data and different attack and failure scenarios. By applying these methods to the same model as used in this study, a complementary perspective can be received, and results can be compared. Thus, a further corroboration of the graph theoretical approach can be tested. Furthermore, this study has investigated the structural vulnerability focusing on node vulnerability. Another approach could be to test methods for quantifying the line vulnerability by using complex network theory concepts. A comparison of these results and results from power flow analyses, possibly including an increasing degree of penetration of variable renewable energy sources, could be performed for further evaluation of the results.

5. Conclusion

In this study, the structural vulnerability of a transmission grid model representing a simplification of the Nordic transmission grid was investigated using exclusively open-source data. This was done through complex network theory, by quantifying topological characteristics in terms of clustering coefficient and four different centrality measures: betweenness, closeness, degree, and combined centrality. The results show that the weighted combined centrality measure was the best in terms of identifying critical nodes in a power grid from a topological point of view. Furthermore, these nodes greatly coincide geographically with the regions presented by the Swedish TSO Svenska kraftnät, which are to be a part of the major grid expansion project NordSyd. Thus, the relevance of the results and the validity of the method with respect to real transmission grid analyses have been strengthened.

This study indicates that complex network concepts appear to pose a useful and yet not computationally heavy method for structural vulnerability analyses of real transmission grids. Specifically, the proposed weighted and combined centrality measure can be used in this context. The advantages of the proposed method are twofold. Firstly, the method is computationally inexpensive, making it possible to analyse relatively large power grids in a short time with reasonable processor capacity. Secondly, the method is suitable to use without detailed knowledge of system parameters, some of which are often confidential. However, it should be emphasised that the results from this method should not be considered conclusive.

CRediT authorship contribution statement

Samuel Forsberg: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Karin Thomas:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **Mikael Bergkvist:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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